

Peichun Hua

Computer Science and Engineering, School of Data Science

The Chinese University of Hong Kong, Shenzhen

[Github](#) — [Website](#) — [Google Scholar](#) — peichunhua@link.cuhk.edu.cn

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EDUCATION

The Chinese University of Hong Kong , Shenzhen (CUHK-Shenzhen)	September 2022 — June 2026
B.Eng in Computer Science and Engineering, School of Data Science (SDS)	
First Class Honors, Outstanding Graduate	
Overall GPA: 3.87/4.00 — Major GPA: 4.00/4.00 — Rank: 2/132	
University of North Carolina , Chapel Hill	September 2024 - May 2025
Overall GPA: 4.00/4.00	
Exchange Student	Computer Science Department

RESEARCH INTERESTS

- Responsible AI and Adversarial Machine Learning; (ACL 2026 Oral, Work Under Review)
- AI and IoT Security: secure hardware design; ML model IP protection (ACSAC 2025, ISCAS 2026, SMC 2026, TODAES Under Review)
- Privacy-Enhancing Technologies: the application of differential privacy (DP), homomorphic encryption (HE), secure multi-party computation (MPC), especially in AI systems; (Work Under Review)

PUBLICATION

* denotes co-first authors of equal contribution.

Under Peer Review

Junguang Yao, **Peichun Hua**, Ying Tao, Yue Zheng. “ReFuseGait: Reliability-Aware Event-Gray Fusion for Gait Recognition.” *Under Review, IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*.

Peichun Hua, Danyang Chen, Junan Zhang, Haifeng Sun, Jingyu Wang, Diwen Xue, Mingyu Li, Yunming Xiao. “Pointing the Way, Hiding the Destination: Practical Private Dense Retrieval at Scale.” *Under Review (Paper (arXiv))*.

Jingdong Jiang*, **Peichun Hua***, Chip-Hong Chang, Wenye Liu, Yue Zheng. “MEDUSA: Proactive Hardware-Anchored Endogenous Model Obfuscation for Ownership-Preserving Continual Learning at the Edge.” *Under Review*.

Peichun Hua, Haoxuan Xu, Mengyuan Li. “Behavioral Skill Reconstruction: Reconstructing Hidden Functionality from LLM Agent Skills.” *Under Review (Paper (arXiv))*.

Peichun Hua, Yunming Xiao. “Exploring Text RAG at Web Scale: Evidence and Lessons from PixelRAG.” *Under Review*.

Peichun Hua, Yunming Xiao. “Matryoshka Hash Representations for Model-Aware Compact Semantic Retrieval.” *Under Review. (Paper (arXiv))*.

Peichun Hua, Yunming Xiao. “Semantics Delivery Network: Rethinking Web Retrieval Infrastructure for LLM Agents.” *Under Review*.

Wentao Du, Siyu Wang, **Peichun Hua**, Linden Chen, Ram Sundara Ramen, Yunming Xiao, Diwen Xue. “The Limits of Passive Network-Stack Fingerprinting in Web Fraud Detection.” *Under Review*.

Peichun Hua, Yunming Xiao. “SPRUCE: Scalable Private Outsourced Retrieval Using Compact Embeddings” *Under Review. (Paper (arXiv))*

Peichun Hua, Yue Zheng. “ChaosFormer: Hardware-Bound Edge Transformer Model Obfuscation for Intellectual Property Protection.” *Under Major Revision, ACM Transactions on Design Automation of Electronic Systems (TODAES), (Paper (PDF))*.

Yanchuan Tang, Junnan Li, Zhan Cheng, **Peichun Hua**, Skylar Zhai, TianMing Sha, Yuan Gao, Ruixiang Tang, Yuanyuan Lei. “MAGMA: Defending Federated Learning via Intrinsic Manifold Geometry.” *Under Review*.

Published / To Appear

Junguang Yao, **Peichun Hua**, Yue Zheng. “Enroll Once, Identify Everywhere: Few-Shot Cross-Scenario Visual Speaker Recognition via Event-driven SlowFast Networks.” To appear at the Pacific Rim International Conference on Artificial Intelligence ([PRICAI 2026](#)) (Oral Presentation, CORE: B, AR = 542/1825 ≈ 29%).

Danyang Chen, Yufeng Gu, Yibo Huang, Chengxuan Pei, **Peichun Hua**, Yang Zhou, Yunming Xiao. “Unlocking Software-defined GPU Fabric Scheduling in the LLM Era.” To appear at the 17th ACM SIGOPS Asia-Pacific Workshop on Systems ([APSys 2026](#)) (AR = 19/43 ≈ 44%).

Xin Wang, **Peichun Hua**, Chip Hong Chang, Wenye Liu, Yue Zheng. “A Unified Open-Set Framework for Scalable PUF-Based Authentication of Heterogeneous IoT Devices.” To appear at the IEEE International Conference on Systems, Man, and Cybernetics ([SMC 2026](#)) (CORE: B, [Code](#), [Paper](#), [PDF](#)).

Peichun Hua, Hao Li, Shanghao Shi, Zhiyuan Yu, Ning Zhang. “Rethinking Jailbreak Detection of Large Vision Language Models with Representational Contrastive Scoring.” In the Annual Meeting of the Association for Computational Linguistics ([ACL 2026](#)). (Main Conference, **Oral Presentation**, CORE: A*, AR = ~2300/12148 ≈ 19%, [Code](#), [Paper](#), [PDF](#), [Slides](#)).

Peichun Hua, Hanxiu Zhang, Tuo Li, Yue Zheng, Wenye Liu. “Live Demonstration: Hardware-Bound IP Protection for OWL-ViT on Edge Devices.” In the IEEE International Symposium on Circuits and Systems ([ISCAS 2026](#)) ([Code](#), [Paper](#), [PDF](#)).

Peichun Hua, Hanxiu Zhang, Tuo Li, Yue Zheng. “Securing On-device Transformer with Hardware Binding and Reversible Obfuscation.” In the Annual Computer Security Applications Conference ([ACSAC 2025](#)) (CORE: A, AR = 84/446 $\approx$ 18.8%, [Code](#), [Paper](#), [Presentation](#), [Slides](#)).

AWARDS

Academic Honors

- **Outstanding Graduate**, School of Data Science, CUHK-Shenzhen 2026
- Dean’s List, School of Data Science, CUHK-Shenzhen 2022–2023, 2023–2024
- Academic Performance Scholarship (Class B, Top <2%), School of Data Science 2023–2024
- Academic Performance Scholarship (Class C, Top <5%), School of Data Science 2022–2023
- University Research Award (with scholarship) Fall 2024, Spring 2025

Research Awards

- *Student Confereceship*, 41st Annual Computer Security Applications Conference, [ACSAC 2025](#)
- *Artifact Available*, *Artifact Reviewed*, and *Artifact Reproducible* Badges — IEEE, for the paper “Securing On-device Transformer with Hardware Binding and Reversible Obfuscation” at [ACSAC 2025](#)
- International Student Research Internship Program (Fully Funded), McKelvey School of Engineering, Washington University in St. Louis Summer 2025

EXPERIENCE

University of Southern California

Los Angeles, CA, USA

Research Assistant (Remote)

May 2026 – Present

- Advisor: [Prof. Mengyuan Li](#)
- Topic: Agentic AI Security. Work completed independently and under review by ACL Rolling Review (ARR).

School of Data Science, CUHK-Shenzhen

Shenzhen, China

Research Assistant

September 2025 – Present

- Advisor: [Prof. Yunming Xiao](#)
- Topics: 1) The security and privacy of RAG pipeline; 2) The application of deep hashing in information retrieval; 3) The infrastructure and performance of RAG in the real web.
- Five papers completed independently and under review. Two full-length papers at system/security conferences, one full-length papers at an AI conference, one workshop vision paper, and one short paper at an AI conference.

McKelvey School of Engineering, Washington University in St. Louis

Missouri, MO, USA

Summer Research Intern

May 2025 – September 2025

- Fully funded by [International Student Research Internship Program](#)
- Advisor: [Prof. Ning Zhang](#)
- Topic: Multi-modal Jailbreak Detection with interpretability methods instantiated with statistical tests and contrastive learning. Achieve highly effective, generalizable, and lightweight attack detection.
- Work completed independently. **Paper accepted by ACL 2026 Oral.**

School of Science and Engineering, CUHK-Shenzhen

Shenzhen, China

Undergraduate Research Intern

June 2024 – May 2025, October 2025 - May 2026

- Advisor: [Prof. Yue Zheng](#)
- Topic: IoT Security, IP (intellectual property) Protection of on-device transformer models.
- Conferred University Research Award (URA, with scholarship) twice. Work completed independently and accepted by ACSAC 2025, IEEE ISCAS 2026, and the extended journal version under major revision at ACM TODAES. One co-first author paper under review at a major security conference.

SKILLS

- **Coursework:** Achieved A grade in all listed *graduate-level* coursework
Cryptography 3D Computer Vision Hardware Security and Side-Channels
Efficient Deep Learning Research Topics in Computer Security Formal Methods in Computer Security
- **Language Proficiency:** TOEFL 108 (R29+L29+S23+W27), GRE 328 (V161+Q167+3.5)

SERVICES

- Reviewer, IEEE/ACM Transactions on Audio, Speech, and Language Processing (TASLP), 2025
- Reviewer, ACL Rolling Review, 2026
- Reviewer, The Pacific Rim International Conference on Artificial Intelligence (PRICAI), 2026